Feedback-Induced Distribution Shift in Multilingual LLM Deployment: Evidence, Characterization, and RL-Based Adaptation Strategies

Mohib Fatima
Independent Researcher | mohib.fatima.edu@gmail.com
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Abstract

Large language models deployed across multilingual production environments often experience performance degradation for underrepresented languages over time. I identify this phenomenon as Feedback-Induced Distribution Shift (FIDS), a systems-level failure in which design and evaluation choices allow localized quality declines to persist while aggregate metrics remain stable.

Drawing on production-scale experience in multilingual LLM deployment, I characterize three recurring patterns through which these failures emerge, give each pattern a minimal formal model with a short proof (elementary results that follow from the stated assumptions), and illustrate the first pattern and the resource-tier gap on public NLLB-200 and FLORES-200 data, where a traffic-weighted aggregate of 57.03 chrF++ sits 43.49 points above the worst-served language.

To address this, I introduce RL-Stable Deployment (RLSD), a proposed inference-time control layer that monitors per-locale quality and adapts prompts, routing, or thresholds without model retraining. RLSD is a design only: it is not implemented, its learning component is not specified, and its worst-case bound is stated as a conjecture with a toy special case.

I argue that these failures arise primarily from system design rather than model limitations, and that addressing them requires new evaluation and deployment paradigms beyond improved benchmarks.

1. Introduction

Over several years of managing multilingual AI products at scale, I repeatedly saw the same pattern: features that worked well in the primary launch markets degraded in underrepresented locales. Aggregate quality metrics looked acceptable, while failures in Tier 2 and Tier 3 markets went undetected. Queries failed to trigger features and outputs were plainly wrong to native speakers, yet none of this showed up in aggregated evaluation.

These failures compound. Users in the affected markets engage less, which reduces feedback signals, which limits opportunities for improvement and widens the gap over time.

The pattern does not require intentional neglect. Metrics are optimized for primary markets, engineering effort follows measurable impact, and low-signal populations are underrepresented in both evaluation and prioritization. The feedback loop exists, and it disadvantages the users who experience the poorest performance.

This paper does not offer a closed solution. It aims to formalize a class of failures that are widely observed in practice but poorly captured by existing evaluation practice.

I call this class Feedback-Induced Distribution Shift (FIDS). Rather than being driven only by changes in data or model updates, FIDS arises from the interaction between deployed models and the feedback systems around them: evaluation metrics, prioritization processes, and organizational incentives. I

separate it into three subtypes, each defined formally in Section 3. The Aggregate Masking Effect (AME) is the case where a traffic-weighted quality metric hides a collapse in a small locale. Low-Resource Locale Collapse (LRLC) is the case where lower quality in a locale reduces human feedback and raises the share of model-generated data, which lowers quality further. Aspirational Deferral Lock (ADL) is the case where fixes for locales with no weight in the prioritization objective are never selected. For two of the three subtypes (AME and ADL) the distribution shifts without any change to the model. LRLC does involve retraining on changing data mixtures, and I model it as such.

FIDS is related to, but distinct from, prior work on data feedback loops (e.g., Taori and Hashimoto, 2023), which focuses on how model outputs influence future training data. In contrast, FIDS operates at the level of deployment and system design, and two of its three subtypes (AME and ADL) require no model retraining at all. The third (LRLC) is closest to the data feedback loop literature. FIDS is also related to performative prediction (Perdomo et al., 2020), which studies distributions that change in response to a deployed model. LRLC can be read as a performative effect of that kind; what FIDS adds is the locale-level view and the evaluation and prioritization channels (AME and ADL), which performative prediction does not model.

This paper makes four contributions. First, a taxonomy of FIDS with three subtypes, each grounded in recurring operational patterns. Second, for each subtype, a minimal formal model and a short proof. These are elementary results (a weighted-average bound, a linear recursion, and a dominance argument); their value is in making the structural claim precise, not in technical depth. Third, an empirical illustration of AME and of the resource-tier gap using public NLLB-200 and FLORES-200 data (Section 6). Fourth, RLSD, a proposed inference-time control layer, presented as a design with a conjectured bound and not as a validated method.

Rather than presenting a fully closed solution, this work aims to formalize a class of failures that are widely observed in practice but insufficiently captured by existing evaluation paradigms.

2. Background and Related Work

2.1 Tiered Deployment and Its Quality Consequences

Multilingual AI systems are commonly deployed using a tiered market strategy, in which initial releases prioritize a small set of high-engagement markets (e.g., the United States, United Kingdom, and other economically dominant regions). These markets typically receive the most comprehensive feature coverage, engineering investment, and pre-launch quality validation. Subsequent deployments extend to lower-tier markets with reduced validation and, in many cases, adaptations of existing features rather than fully localized implementations. While this approach reflects a practical allocation of limited resources, it introduces systematic disparities in product quality that are often insufficiently measured or tracked over time.

A key consequence of tiered deployment is that features deployed in lower-tier markets are frequently calibrated to the query distributions and usage patterns of higher-tier markets. As a result, they may fail to generalize to local linguistic structures, cultural contexts, or region-specific user intents. For example, a feature that reliably triggers for English queries with relatively fixed syntactic patterns may exhibit degraded or inconsistent behavior when exposed to morphologically richer or less standardized language forms.

These failures are not necessarily indicative of deficiencies in the underlying model. Rather, they reflect a mismatch between the conditions under which the system was developed and the environments in which it is deployed. In this sense, tiered deployment introduces a form of deployment-induced error, where

system performance degrades due to insufficient adaptation to local contexts rather than inherent model limitations.

2.2 Distributionally Robust Machine Learning

Prior work in distributionally robust machine learning has highlighted the limitations of optimizing for average-case performance in heterogeneous data settings. Sagawa et al. (2020) show that models trained using standard empirical risk minimization can achieve strong aggregate performance while exhibiting significantly worse outcomes on underrepresented subgroups. Similarly, Hashimoto et al. (2018) propose a distributionally robust optimization (DRO) framework that improves worst-case subgroup performance without requiring explicit group annotations, and Oren et al. (2019) apply the same idea to language modeling.

This line of work is directly relevant to the tiered deployment dynamics described in Section 2.1, as both highlight systematic disparities that are obscured by aggregate evaluation metrics. However, the focus of these approaches is primarily on mitigating bias during training by adjusting optimization objectives or data weighting strategies.

In contrast, the phenomenon described in this paper, Feedback-Induced Distribution Shift (FIDS), emerges after deployment. It arises not from the training distribution itself, but from the interaction between a deployed model and the feedback systems that govern its evaluation and iteration. As such, FIDS reflects a form of deployment-time bias that compounds through measurement practices, prioritization decisions, and uneven feedback signals, rather than through the model's learned parameters alone.

2.3 LLM Evaluation Frameworks

Liang et al. (2022) introduce HELM (Holistic Evaluation of Language Models), a comprehensive framework for evaluating large language models across a range of scenarios, metrics, and user populations. HELM's multilingual evaluations highlight systematic performance differences across languages and task settings, providing an important baseline for understanding cross-lingual variability in model behavior.

However, HELM evaluates model performance at a fixed point in time under controlled and standardized conditions. As a result, it does not capture how performance evolves after deployment, particularly in production environments where evaluation, prioritization, and iteration are shaped by real-world usage patterns and organizational constraints.

The phenomenon described in this paper builds directly on these observations. While HELM surfaces cross-lingual performance disparities at evaluation time, Feedback-Induced Distribution Shift (FIDS) characterizes how such disparities can persist, amplify, or become less visible over the operational lifetime of a system. In this sense, static evaluation frameworks provide an initial snapshot of quality differences, whereas FIDS focuses on their dynamic evolution under deployment conditions.

2.4 Organizational Dynamics in AI Development

The influence of organizational incentives on AI system quality remains relatively underexplored in the technical literature, despite its practical importance in large-scale deployments. Prior work on machine learning systems has examined forms of technical debt that arise from increasing system complexity and maintenance burdens. For example, Sculley et al. (2015) show how design and engineering decisions introduce long-term maintenance costs that are not visible at deployment time.

However, these frameworks primarily focus on code, data pipelines, and model behavior, and do not explicitly address how organizational priorities and incentive structures shape the distribution of system quality across different user populations. In practice, factors such as feature velocity, market prioritization, and competitive pressure can influence which issues are addressed and which are deferred, potentially leading to uneven quality outcomes across regions and languages.

The framework introduced in this paper builds on this gap by examining how such dynamics affect multilingual deployments in particular. While prior work has not, to my knowledge, formally characterized these effects in this context, the Aspirational Deferral Lock (ADL) subtype proposed later in this paper provides one mechanism, stated as a proposition with a short proof in Section 3.4, through which organizational incentives can contribute to persistent and compounding quality disparities.

2.5 Language Resource Tiers

Joshi et al. (2020) classify the world's languages into six classes by the availability of labeled and unlabeled data, and show that NLP resources are heavily concentrated in a small number of languages. The tier vocabulary used in this paper (Tier 2 and Tier 3 markets) is a market-side analogue of that data-side classification, and Section 6 uses the data-side version to define resource tiers. Large multilingual translation models show the same pattern in practice: the NLLB-200 evaluation (NLLB Team, 2022) covers 200 languages with quality that varies widely across them.

2.6 Aggregate Metrics Hiding Subgroup Harm

Audits of aggregate accuracy hiding subgroup harm predate this paper. Buolamwini and Gebru (2018) found that commercial gender classification systems had error rates of up to 34.7% for darker-skinned women against at most 0.8% for lighter-skinned men, a disparity that aggregate reporting did not reveal. The Aggregate Masking Effect (AME, Section 3.2) is the same arithmetic applied to locales and traffic share. I do not claim it as a new observation; Proposition 1 is a short statement of why it happens whenever a dashboard reports a traffic-weighted average.

2.7 Performative Prediction

Perdomo et al. (2020) formalize settings where the data distribution changes in response to the deployed model. Low-Resource Locale Collapse (LRLC, Section 3.3) is an instance of this pattern: lower quality reduces engagement, which changes the data the system sees. The framing here differs in two ways. It is organized by locale, so the quantity of interest is the worst-served group and not the population average, and it adds two channels that sit outside the data-generating population, namely what the evaluation metric can see (AME) and what the organization chooses to fix (Aspirational Deferral Lock, ADL, Section 3.4).

3. A Taxonomy of Feedback-Induced Distribution Shift

Feedback-Induced Distribution Shift (FIDS) can be decomposed into three recurring subtypes, each corresponding to a distinct underlying mechanism and observable pattern in deployed systems. These subtypes differ in how quality degradation emerges, how it manifests in production telemetry, and how it must be detected and mitigated in practice.

While analytically separable, the subtypes are not independent. In real-world deployments, they often co-occur and interact, leading to compounding effects that amplify disparities across user populations. This interaction is a key feature of FIDS: localized failures are not isolated, but can reinforce one another through shared feedback and prioritization dynamics.

The taxonomy presented here is derived from recurring patterns observed across multilingual deployments. Below, each subtype is given both an operational description and a minimal formal model with a short proposition, so that the claim "this failure is structural" is a mathematical statement under stated assumptions and not only an inference from anecdote.

3.1 Setup and Notation

Let there be L locales, indexed i = 1, …, L. At time t, let $n_i(t)$ denote query volume (traffic) from locale i, and let $q_i(t) \in [0,1]$ denote the quality of the deployed system for locale i (accuracy, triggering precision, or any bounded quality metric). Let $N(t) = \Sigma_i\, n_i(t)$ be total traffic. The aggregate metric reported on dashboards is the traffic-weighted average of per-locale quality:

$$Q_{agg}(t) = (1 / N(t)) \cdot \Sigma_{i=1}^{L}\, n_i(t)\, q_i(t)$$

This is not a simplification of convenience; it is commonly how "topline" quality metrics are computed in production systems, because raw per-query accuracy is what is measured, and it gets rolled up by summing over queries. The three propositions below all take Q_{agg}, or a component of the system that produces it, as their object of study.

3.2 Aggregate Masking Effect (AME)

The Aggregate Masking Effect (AME) arises when system performance is monitored using aggregate metrics that obscure disparities across user populations. Because these metrics are typically weighted by traffic volume, they disproportionately reflect the performance experienced by majority groups, while masking degradation in lower-volume populations.

The mechanism follows directly from the structure of weighted aggregation. Consider a deployment in which a dominant market accounts for the majority of traffic, while smaller markets contribute a relatively small fraction. An aggregate metric, such as accuracy, engagement, or user satisfaction, will be dominated by the performance of the highest-weight group. As a result, substantial quality gaps in lower-weight groups may have only a negligible effect on the reported metric. For example, a system that achieves 94% accuracy for a high-traffic group and 61% accuracy for a lower-traffic group may still report aggregate accuracy above 90% (this holds whenever the lower-traffic group carries under about 12% of queries), depending on the traffic distribution.

Proposition 1 (Aggregate Masking Effect). An arbitrarily severe quality collapse in a subset of locales can produce an arbitrarily small change in the reported aggregate metric, provided that subset's traffic share is small enough.

Setup. Hold traffic n_i fixed. Suppose a subset S of locales degrades: $q_i \to q_i - \delta_i$ for $i \in S$, with $0 \le \delta_i \le \min(\delta_{max}, q_i)$ so that quality stays in [0,1], while all other locales are unchanged. Let $w_S = (1/N)\, \Sigma_{i \in S}\, n_i$ be the traffic share of the degraded locales.

$$Q_{agg}^{\,before} - Q_{agg}^{\,after} = (1/N)\, \Sigma_{i \in S}\, n_i\, \delta_i \le \delta_{max} \cdot w_S$$

Proof. Direct substitution: only terms for $i \in S$ change, each by at most $n_i \cdot \delta_{max}$, so the total change is at most $\delta_{max} \cdot \Sigma_{i \in S}\, n_i = \delta_{max} \cdot N \cdot w_S$; dividing by N gives the bound. ■

Why this matters. If a monitoring system alerts only when Q_{agg} drops below threshold τ, Proposition 1 shows the alert cannot fire (starting from $Q_{agg}^{\,before} > \tau$) whenever $\delta_{max} \cdot w_S < Q_{agg}^{\,before} - \tau$, that is, whenever $w_S < (Q_{agg}^{\,before} - \tau) / \delta_{max}$, regardless of how severe the underlying degradation is within each affected locale (δ_i can be as large as q_i). Masking is not a monitoring bug to be tuned away; it follows directly from using a traffic-weighted average as the sole detection signal. In practice, this results in a failure mode where system-level metrics indicate acceptable performance while meaningful segments of the user

population experience degraded or inconsistent behavior: reduced feature activation rates, misalignment with local linguistic patterns, or degraded response quality, none of which are captured by aggregate reporting alone.

Mitigation strategies, such as disaggregated evaluation and locale-specific alerting, are well understood in principle. However, in practice, they are often underutilized, in part because surfacing fine-grained performance gaps introduces additional prioritization and engineering overhead, a dynamic formalized separately as Aspirational Deferral Lock in Section 3.4.

3.3 Low-Resource Locale Collapse (LRLC)

Low-Resource Locale Collapse (LRLC) describes a self-reinforcing degradation process that can emerge once performance disparities are no longer visible in aggregate monitoring. Building on the masking effect described in Section 3.2, LRLC captures how unaddressed quality gaps in lower-resource locales can compound over time through feedback-driven dynamics.

The core mechanism is a negative feedback loop between system quality and user engagement. When performance in a given locale degrades, users are less likely to engage with the affected features. Reduced engagement leads to diminished data signals, such as fewer interactions, lower-quality feedback, or reduced coverage of relevant query patterns, which in turn limits opportunities for system improvement. This reduced improvement capacity allows performance gaps to persist or widen, further suppressing engagement.

This process produces a characteristic signature in production data. Over time, query volume and feature usage in affected locales may decline, either because users disengage entirely or adapt their behavior to avoid unreliable functionality. As a result, the system receives progressively less informative signal from these populations. In many cases, by the time degradation becomes visible in aggregate metrics, the affected user base has already contracted substantially.

Proposition 2 (Low-Resource Locale Collapse). Under the model below, locales with a larger model-generated share of training data settle at a lower quality than locales with a smaller share, even when both start from similar quality.

Setup. Let training data for locale i at time t+1 be a mixture of human-verified examples (quality q_{human}, fixed) and model-generated examples, with mixture weight $\alpha_i(t) \in [0,1]$ on the model-generated share. In practice α_i rises as human annotation capacity for locale i falls; this is the discrete-time analogue of the continuous decline in q(t) described operationally above. Following the model-collapse literature (Shumailov et al., 2023), assume model-generated examples have expected quality $q_i(t) - \varepsilon$ for some $\varepsilon \geq 0$ reflecting compounding self-reference. Quality is bounded, so the recursion is clipped to [0,1] (Section 3.1). The proposition assumes that α_i is higher in low-resource locales; it does not derive that, and shows what follows from it. Then:

$$q_i(t+1) = clip_{[0,1]}((1 - \alpha_i)\, q_{human} + \alpha_i\, (q_i(t) - \varepsilon))$$

Result. Without clipping, this is a linear recursion in $q_i(t)$ with fixed point $q_i^* = q_{human} - [\alpha_i/(1-\alpha_i)]\, \varepsilon$ when $\alpha_i < 1$, approached geometrically at rate α_i. Because quality cannot go below 0, the fixed point of the clipped recursion is $\max(0, q_i^*)$. It reaches 0 exactly when $\alpha_i \geq q_{human} / (q_{human} + \varepsilon)$. As $\alpha_i \to 1$ (locale almost entirely dependent on model-generated data), the settled quality falls toward 0 for any $\varepsilon > 0$; the well-resourced case ($\alpha_i \to 0$) stays anchored near q_{human} regardless of ε.

$$q_i^* = q_{human} - [\alpha_i / (1 - \alpha_i)]\, \varepsilon$$

Proof. Standard linear recursion analysis: $q_i(t+1) = \alpha_i\, q_i(t) + [(1-\alpha_i)\, q_{human} - \alpha_i \varepsilon]$ has the fixed point above, and converges to it since $0 \le \alpha_i < 1$. With clipping at 0, the map $q \mapsto \max(0, \alpha_i\, q + c)$ is still a contraction with factor α_i, so it has a unique fixed point and the iteration converges to it. Write $c = (1-\alpha_i)\, q_{human} - \alpha_i \varepsilon = (1-\alpha_i)\, q_i^*$. If $q_i^* < 0$ then $c < 0$ and 0 is a fixed point of the clipped map; otherwise q_i^* is. The upper bound 1 never binds, since $q_{human} \le 1$ and $\varepsilon \ge 0$ keep every update at or below 1. ■

Why this matters. This gives LRLC a precise mechanism: it is not just "less data, worse model"; it is that the composition of what little data exists shifts toward self-generated content specifically where human annotation investment is weakest, and that shift is what compounds. A defining feature of LRLC is that its early stages are often obscured by the Aggregate Masking Effect (Proposition 1): a locale's contribution to Q_{agg} stays small even as α_i rises, so intervention is delayed until the feedback loop has significantly reduced both q_i and the human-verified share available to correct it, making recovery more difficult. This is also the direct theoretical bridge to the RL-based adaptation strategies of Section 4: under the model, a policy that raises the human-verified share $(1-\alpha_i)$ specifically where α_i is high keeps q_i^* above a target q_{min} whenever $\alpha_i/(1-\alpha_i) \le (q_{human} - q_{min}) / \varepsilon$.

3.4 Aspirational Deferral Lock (ADL)

Aspirational Deferral Lock (ADL) describes the organizational dynamics that prevent the remediation of persistent quality gaps, even after they have been identified. While Aggregate Masking Effect (AME) obscures disparities and Low-Resource Locale Collapse (LRLC) drives their compounding, ADL explains why these issues often remain unresolved in practice.

ADL arises when quality improvement efforts for underrepresented user populations are consistently deprioritized relative to the development of new features. In many large-scale AI deployments, identified issues are placed into backlogs or long-term planning categories without clear ownership, timelines, or allocated resources. Over time, this leads to the accumulation of what can be described as quality debt, unresolved performance deficiencies that disproportionately affect specific user groups.

A distinguishing aspect of ADL is the absence of a concrete resolution pathway. Unlike traditional notions of technical debt, which typically imply eventual remediation, deferred quality issues in this context may persist indefinitely due to shifting priorities and limited engineering capacity. This dynamic is often reinforced in environments where success is measured primarily by feature velocity, product expansion, or competitive positioning, rather than by uniformity of user experience across populations.

Proposition 3 (Aspirational Deferral Lock). A backlog item can have organization-wide consensus that it is real and severe, and still have zero probability of ever being fixed under a greedy resource-allocation rule, if it carries zero weight in the objective function driving that allocation, as long as positive-weight work keeps arriving.

Setup. Model engineering-effort allocation as a greedy scheduler: at each step, one unit of effort goes to the locale maximizing $w_i \cdot g_i$, where $g_i \ge 0$ is the marginal quality gain from fixing locale i's backlog item (the "everyone agrees it's real" term), and $w_i \ge 0$ is that locale's weight in the organization's objective, in practice, a function of which team's OKRs or traffic tier the locale falls under.

$$argmax_i\, w_i \cdot g_i$$

Result. If $w_i = 0$ for locale i, then locale i is never selected by this rule in any round in which some other locale i' has $w_{i'} \cdot g_{i'} > 0$, regardless of g_i.

Proof. For any i with $w_i = 0$: $w_i \cdot g_i = 0 < w_{i'} \cdot g_{i'}$ for any i' with $w_{i'} > 0$ and $g_{i'} > 0$, so i is strictly dominated in every such round. ■

Scope. The result is conditional. It holds only while some positive-product item remains in the backlog; if the positive-weight work were ever exhausted, a zero-product item could be selected. With finite effort, the claim is therefore that a zero-weight item waits behind all positive-weight work, and it is starved in practice when new positive-weight work keeps arriving (new features, new launches), which is the pattern Section 7.2 describes.

Why this matters. This is deliberately a simple statement, because the phenomenon is simple: the "aspirational backlog" is not a prioritization failure in the sense of bad judgment; it is what greedy allocation under a zero-weight objective always produces, as a matter of the algorithm's structure, independent of how large g_i is. In this model, a zero-weight item is never resolved while positive-weight work keeps arriving, however large g_i is. The implied fix is direct: ADL only breaks if w_i is made a function of backlog age or locale coverage rather than current traffic, exactly the kind of policy-level RL adaptation strategy explored in Section 4.

The interaction between ADL and LRLC is particularly consequential. As performance degrades in a given locale and engagement declines (Proposition 2), the resulting reduction in user activity further lowers n_i and hence w_i in most real objective functions, which by Proposition 3 further reduces the probability that locale is ever selected for remediation. This creates a self-reinforcing cycle in which the populations experiencing the lowest quality also generate the weakest signals for intervention.

3.5 Connection to Section 4

The three propositions above are exactly what the RLSD framework in Section 4 is designed to act against: Proposition 1 predicts when a monitoring threshold fails to fire; Proposition 2 predicts the quality trajectory under different α_i regimes and the fixed point it converges to; Proposition 3 predicts which backlog items a greedy scheduler will starve. RLSD's three interventions in Section 4.2 map directly onto these three failure modes.

4. RL-Stable Deployment (RLSD): An Inference-Time Adaptation Framework

4.1 Why Inference-Time Adaptation

The failure modes described in Section 3 arise from a combination of technical and organizational factors. In principle, they can be addressed through improved monitoring practices, targeted investment in underrepresented locales, and organizational prioritization of quality alongside feature development. However, such interventions often require sustained coordination, resource allocation, and changes in incentive structures, which may be difficult to implement in practice, particularly in fast-moving or resource-constrained environments.

RL-Stable Deployment (RLSD) is designed as a complementary, system-level mitigation that operates within these constraints. Rather than relying on model retraining or organizational restructuring, RLSD introduces an inference-time adaptation layer that adjusts model behavior in response to signals derived from production telemetry.

At a high level, RLSD is a controller that takes feedback signals as input, such as discrepancies in engagement, feature activation rates, or performance proxies across user populations, and selects inference-time interventions. A learned policy (for example, one trained with reinforcement learning) is the intended long-term form, but this paper does not specify or train one. The goal is to reduce disparities that would otherwise persist undetected or unaddressed.

Importantly, RLSD does not eliminate the underlying causes of FIDS. Instead, it provides a mechanism for mitigating their downstream effects in deployed systems, particularly in settings where more direct interventions are not immediately feasible.

4.2 Framework Design

RL-Stable Deployment (RLSD) introduces a lightweight control layer that operates alongside a deployed language model and continuously ingests production telemetry. This telemetry includes locale-disaggregated query distributions, concretely, per-locale $n_i(t)$ and estimates of $q_i(t)$ as defined in Section 3.1, engagement rates, feature activation statistics, and available proxies for user satisfaction. Rather than relying on the aggregate Q_{agg}, RLSD maintains a structured view of system performance across user populations over a rolling time window.

The RLSD controller can be formulated as a contextual decision process. At each timestep, the controller observes a state s_t composed of telemetry-derived signals (e.g., per-locale engagement trends, trigger rates, and relative performance disparities). Based on this state, it selects an action a_t from a constrained set of inference-time interventions. The objective is to optimize a reward signal that balances overall system performance with penalties for cross-locale disparities, thereby encouraging more uniform quality across populations. One candidate reward, not evaluated here, is $r_t = (1 - \lambda)\, Q_{agg}(t) + \lambda\, W(t)$, where $W(t)$ is the worst-locale quality defined in Section 4.3 and $\lambda \in [0,1]$ sets how strongly the controller trades aggregate quality for worst-case quality. No learning algorithm, training procedure, or experiment is provided. A simple threshold rule that triggers an intervention when $W(t)$ falls below a floor is the natural non-learned baseline.

These interventions are designed to be lightweight and composable within existing inference pipelines. They include:

- Prompt adaptation: applying locale-specific prefixes or instruction tuning at inference time
- Routing strategies: directing queries to specialized model variants or calibrated pipelines
- Threshold adjustments: modifying feature triggering conditions based on observed locale behavior

The mapping between FIDS subtypes and interventions follows directly from the propositions in Section 3:

AME (Proposition 1): RLSD monitors disaggregated $q_i(t)$ rather than only Q_{agg}, and identifies cases where a locale's degradation δ_i is large even though w_S is too small for it to move Q_{agg} past threshold τ. When such discrepancies are detected, the controller prioritizes targeted interventions, localized prompt adjustments or routing, for the affected group without altering global behavior.

LRLC (Proposition 2): RLSD tracks the effective mixture weight α_i implied by observed engagement and annotation trends. Sustained increases in α_i, particularly when correlated with degraded q_i, are treated as indicators of an approaching fixed point q_i^* below an acceptable bound. In response, RLSD increases the effective representation of these locales within the inference pipeline and triggers auxiliary processes such as data augmentation or targeted evaluation, interventions that act directly to lower α_i, the quantity Proposition 2 identifies as the lever.

ADL (Proposition 3): Because ADL originates from organizational dynamics external to the model, RLSD cannot alter w_i directly. Instead it focuses on improving visibility: the system generates structured reports that quantify persistent quality gaps across locales, in effect, estimates of g_i for every locale with $w_i = 0$, including their estimated impact on engagement and retention. By translating quality disparities into operational metrics, these reports aim to reduce the likelihood that such issues remain indefinitely deprioritized.

Overall, RLSD does not replace existing training or evaluation workflows. Instead, it augments deployed systems with a feedback-sensitive control layer that enables localized adaptation and earlier detection of emerging disparities.

4.3 Theoretical Properties

RLSD is intended to bound worst-case performance across user populations under the failure modes described in Section 3. The bound below is a conjecture, with one toy special case proved. Let $W(t) = \min_i q_i(t)$ denote the minimum performance achieved across all locale groups at time t.

In the absence of targeted intervention, the combined effects of AME (Proposition 1), LRLC (Proposition 2), and ADL (Proposition 3) can lead to sustained degradation in worst-case performance: AME allows a locale to degrade without tripping an aggregate alert, LRLC gives that degradation a self-reinforcing trajectory toward $q_i^* < q_{human}$, and ADL leaves the responsible backlog item unselected while positive-weight work remains. In particular, when feedback signals from low-resource locales are weak or systematically deprioritized, performance in these groups may decline over time and fail to recover without external correction.

RLSD mitigates this behavior by introducing periodic monitoring and adaptive intervention. Under the assumption that (i) degradation in any locale is detectable through telemetry signals with bounded delay, and (ii) the intervention set includes actions capable of partially correcting observed disparities, I conjecture that the system can maintain a lower bound on worst-case performance. Formally, for monitoring frequency f, the conjecture is that

$$W(t) \geq W(0) - \delta(f)$$

where δ(f) is a function that captures the maximum allowable degradation between successive interventions, non-increasing in f. As the monitoring frequency increases, the time between corrective actions decreases, leading to a tighter bound on worst-case performance.

Toy special case (Proposition 4). Suppose the worst-locale quality W(t) can fall by at most d per unit time between interventions, and suppose interventions occur at times $t_k = k/f$, each restoring W to at least its value just after the previous intervention (with W(0) as the starting value). Then $W(t) \geq W(0) - d/f$ for all t, that is, $\delta(f) = d/f$.

Proof. Write $W(t_k^+)$ for the value just after the k-th intervention. By the restoration assumption, $W(t_k^+) \geq W(t_{k-1}^+)$, so by induction $W(t_k^+) \geq W(0)$ for every k. For t in $[t_k, t_{k+1})$, $W(t) \geq W(t_k^+) - d\,(t - t_k) \geq W(0) - d/f$. ■

This is elementary and it moves the difficulty without removing it. The open question is whether inference-time actions (prompts, routing, thresholds) can deliver the restoration assumption for a given locale and failure mode, and at what detection delay. I do not claim they can.

This guarantee is inherently conditional: it depends on the quality of telemetry signals, the responsiveness of the intervention mechanism, and the effectiveness of available actions. Unlike Propositions 1–3, the general bound is not proved from a fully specified model; closing that gap (deriving δ(f) explicitly as a function of detection-delay and intervention-strength assumptions weaker than the toy case) is identified as future work in Section 7.3. Nevertheless, it provides a useful operational target: sufficiently frequent and targeted intervention is conjectured to prevent unbounded degradation in underrepresented populations.

Importantly, the bound is defined over worst-case locale performance rather than aggregate metrics. This distinction is central to the FIDS framework: maintaining high average performance does not preclude

severe degradation in specific user groups. By focusing on the minimum performance across populations, RLSD directly targets the failure modes that aggregate evaluation obscures.

5. Why Manual Workarounds Are Not Enough

Prior to the development of a formal framework, mitigation efforts for multilingual quality gaps often relied on manual, localized interventions. These included prompt-level adjustments, tuning of feature triggering thresholds, and calibration of precision–recall tradeoffs to better align with the requirements of specific markets. In many cases, such interventions were effective in improving performance for individual features and locales.

However, these approaches exhibit fundamental limitations. First, they are inherently reactive: they depend on the prior identification of a problem, which in turn requires access to disaggregated performance data that is not always available in standard monitoring pipelines, precisely the data Proposition 1 shows is necessary, since Q_{agg} alone cannot surface it. Second, they are localized and non-transferable; each intervention addresses a specific feature–locale combination without providing a general mechanism for detecting or mitigating similar issues elsewhere. Third, they lack persistence. Without continuous maintenance, improvements degrade over time as systems evolve and underlying conditions shift.

These limitations make manual approaches difficult to scale in large, multilingual deployments. Even when effective in isolation, they do not provide a systematic way to monitor, prioritize, and sustain quality improvements across diverse user populations.

RL-Stable Deployment (RLSD) is designed to address this gap. Rather than replacing expert-driven interventions, RLSD provides the infrastructure required to make such interventions consistent and scalable. By enabling continuous detection of performance disparities and supporting targeted, inference-time adaptation, RLSD transforms what would otherwise be ad hoc remediation into a structured, system-level process.

6. Empirical Illustration

Sections 3 and 4 are theory and design. This section checks two of the claims against public data. It is an illustration, not a controlled study, and it does not test RLSD or the feedback dynamics of LRLC.

6.1 Data and Setup

Model and benchmark. I use NLLB-200 (NLLB Team, 2022), the 3.3B dense variant, evaluated on FLORES-200 devtest, with chrF++ (Popović, 2017) as the quality metric. Per-direction chrF++ scores come from the public release that accompanies Koishekenov et al. (2023) (github.com/naver/nllb-pruning), which includes scores for the dense 3.3B baseline in all 202 × 201 = 40,602 translation directions. For each of the 201 non-English languages I define q_i as the mean of its English$\rightarrow$X and X$\rightarrow$English chrF++ scores.

Traffic. Real production traffic is not public, so the traffic shares are synthetic and labeled as such. In the primary scheme, fixed before I looked at any result, ten widely spoken languages (Spanish, Mandarin, Portuguese, French, German, Arabic, Russian, Japanese, Hindi, Indonesian) carry 80% of traffic equally, and the remaining 191 languages share the other 20% equally, about 0.105% each. I also ran three other schemes (a Zipf split within the ten, and 60% and 90% of traffic on the ten) and report the range. These are my assumptions, not measurements.

Resource tiers. I use the language classes of Joshi et al. (2020), from the taxonomy file released with that paper. I matched FLORES language names to taxonomy names by a fixed rule (with a short list of name aliases). A language was excluded if the taxonomy had no entry for it or listed it under more than one class. Of the 201 languages, 175 matched. Fourteen had no entry (mostly Arabic dialects) and twelve were listed under conflicting classes. Tiers were set before the scores were joined: Low is classes 0 and 1, Mid is classes 2 and 3, and High is classes 4 and 5.

Release. The scripts that produce every number below, and the computed results, are in the empirical/ folder of the companion repository (github.com/mohib-fatima/fids-llm-deployment). The scores come from a source released under CC BY-NC-SA 4.0, so the per-language table in that folder carries the same terms.

6.2 Aggregate Masking

Quantity	Value
Traffic-weighted aggregate, Q_{agg} (primary scheme)	57.03 chrF++ (55.19 to 57.96 across schemes)
Unweighted mean across 201 languages	50.11 chrF++
Worst-served language (Kanuri, Arabic script)	13.54 chrF++
Traffic share of that language	0.105%
Gap between aggregate and worst language	43.49 chrF++ points (41.64 to 44.41 across schemes)
Ten lowest-scoring languages: combined traffic share	1.05%

A dashboard reporting the traffic-weighted aggregate shows 57.03 chrF++. The worst-served language scores 13.54 and carries about 0.105% of traffic, so it sits 43.49 points below the number on the dashboard. The ten lowest-scoring languages together carry 1.05% of traffic and average 24.47, less than half the aggregate. Proposition 1 gives the matching bound for a degradation event. If the worst language fell from 13.54 to 0, the aggregate would move by at most $0.105\% \times 13.54 \approx 0.014$ chrF++ points. If all ten of the lowest-scoring languages fell to 0, the aggregate would move by at most 0.26 points. Both are far below any plausible alert threshold. This is the same arithmetic as Proposition 1, applied to a real quality distribution, and the gap stays between 41.6 and 44.4 points under every traffic scheme I tried.

6.3 Resource-Tier Gap

Tier	Languages	Mean chrF++
High-resource (classes 4 and 5)	25	57.27
Mid-resource (classes 2 and 3)	42	56.34
Low-resource (classes 0 and 1)	108	46.48

The High to Low gap is 10.79 points (bootstrap 95% interval 6.49 to 14.74, 10,000 resamples). The Spearman correlation between class and quality across the 175 languages is 0.52 (interval 0.39 to 0.64). By class, the mean is 36.21 for class 0 (19 languages), 48.68 for class 1 (89), 50.94 for class 2 (16), 59.66 for class 3 (26), 59.31 for class 4 (18), and 52.02 for class 5 (7). Quality rises with resource class up to class 3 and is flat above it, and class 0 averages about 12 points below class 1 and more than 20 points below classes 3 and 4. This is a cross-sectional snapshot. It is consistent with the tier pattern that LRLC describes, but it does not show the feedback dynamic itself, which would need longitudinal telemetry.

The High and Mid tiers are close, and class 5 is lower than classes 3 and 4. chrF++ is a character n-gram metric, and Chinese and Japanese score low on it for script-related reasons that are not about resource level; both are in the top classes. Removing Chinese (Simplified and Traditional), Yue Chinese, and Japanese raises the High mean to 59.94 and the High to Low gap to 13.36, which is consistent with that

explanation but does not prove it. I rely on the High versus Low comparison and not on the ordering of High and Mid.

6.4 What This Does and Does Not Show

- The traffic weights are synthetic. The AME gap is real in the quality data, and its size relative to the dashboard number depends on the assumed traffic distribution, though it varies only between 41.6 and 44.4 points across the four schemes I tried.
- The tier comparison covers 175 of 201 languages. Classes come from a data-availability taxonomy from 2020 and were matched to FLORES names by rule, so a small number of language-name matches may be imperfect.
- chrF++ has script-level biases, noted above.
- Both results are snapshots of one model. They show that the phenomena are visible in real quality data. They do not show that every multilingual system behaves this way.
- Nothing here evaluates RLSD.

7. Discussion

7.1 The Measurement Problem

A central implication of the FIDS framework, and of Proposition 1 in particular, is that the associated failure modes are often not captured by commonly used aggregate quality metrics. Standard evaluation practices typically emphasize average performance across users, which can obscure substantial disparities between user populations.

Addressing this issue is not simply a matter of introducing additional metrics into existing dashboards. Rather, it requires a shift in how multilingual AI quality is conceptualized. In particular, evaluation frameworks must account for performance variation across populations, rather than relying solely on aggregate summaries.

This shift has practical consequences. Consider a system that achieves 95% average accuracy while delivering substantially lower performance, e.g., 50% accuracy, for a minority subset of users. Aggregate reporting would characterize this system as high-performing, despite the fact that a nontrivial portion of its user base experiences significantly degraded outcomes.

From the perspective of multilingual system evaluation, such a characterization is incomplete. A more appropriate assessment must consider both overall performance and the distribution of that performance across user groups, including the worst-case outcomes W(t) introduced in Section 4.3. Without this perspective, evaluation metrics risk overstating system quality and obscuring disparities that are central to the failures described by FIDS.

7.2 The Novelty Trap

Proposition 3 (Aspirational Deferral Lock) highlights the role of organizational incentive structures in shaping how AI system improvements are distributed across user populations. In many competitive development environments, success is primarily measured through the introduction of new capabilities, user growth, and market differentiation. As a result, engineering effort tends to be directed toward feature expansion rather than systematic improvement of existing functionality.

This dynamic has uneven consequences. Efforts to improve quality for underrepresented user groups, such as speakers of lower-resource languages or users in lower-tier markets, often compete with initiatives that have more immediate and visible impact on headline metrics. Because these populations

may contribute less to aggregate engagement or revenue signals, issues affecting them are less likely to be prioritized, even when they are known to be real, as Proposition 3 predicts for any locale with $w_i = 0$ while positive-weight work keeps arriving.

Importantly, this pattern does not depend on individual intent. It emerges from the alignment between organizational incentives and the metrics used to evaluate success. When these metrics do not adequately capture disparities in user experience, optimization processes can systematically overlook segments of the population that fall outside dominant usage patterns.

Within this context, FIDS can be understood as a consequence of optimizing a complex sociotechnical system under incomplete measurement. When key aspects of user experience are not represented in the objective function, whether explicitly or implicitly, feedback loops may reinforce existing disparities rather than reduce them.

7.3 Limitations and Future Work

This work has several limitations that point to important directions for future research.

First, the proposed FIDS taxonomy (AME, LRLC, ADL) is derived from observed patterns in large-scale deployments rather than from a controlled empirical study spanning multiple organizations or system architectures. Propositions 1–3 establish that each mechanism is structurally capable of producing the described failure under the stated assumptions; they do not establish how often, or how severely, each mechanism dominates in any particular deployment. The illustration in Section 6 uses public benchmark data for one model and synthetic traffic weights. It shows that the AME arithmetic and the tier gap are visible in real quality data; it does not measure how often either mechanism dominates in production. The taxonomy should be viewed as a conceptual framework, formally grounded but requiring further empirical validation across diverse deployment contexts.

Second, RL-Stable Deployment (RLSD) is presented as a system design and has not been implemented or evaluated in a production environment. The bound in Section 4.3 is a conjecture, with a toy special case proved; deriving $\delta(f)$ under weaker assumptions, and characterizing the conditions under which the detectability and intervention-effectiveness assumptions hold, is left to future work.

Third, the proposed approach to addressing Aspirational Deferral Lock (ADL) emphasizes improved measurement and visibility through mechanisms such as locale-level quality reporting. Proposition 3 shows that a nonzero w_i is necessary for a locale to ever be selected under greedy allocation; it does not show that visibility alone is sufficient to make an organization set $w_i > 0$. The relationship between measurement and organizational action is complex, and increased visibility alone may not be sufficient to drive sustained remediation.

Fourth, Proposition 1 treats traffic shares n_i as fixed, whereas LRLC implies that traffic in an affected locale may itself decline as quality falls. I analyze the two separately and do not model their interaction.

Future work should address these limitations along several dimensions. Empirically, the FIDS framework can be evaluated by analyzing multilingual system performance across multiple deployment settings to assess the prevalence and interaction of the proposed subtypes. From a systems perspective, implementing RLSD and measuring its impact on both aggregate and worst-case performance would provide critical validation of its practical utility. Finally, further research is needed to understand how organizational structures and incentive mechanisms can be aligned with quality objectives in multilingual systems.

8. Conclusion

This paper is motivated by a recurring pattern observed in large-scale multilingual AI deployments: systems that appear to perform well according to aggregate metrics can simultaneously provide significantly degraded experiences for specific user populations. These failures are often not visible in standard evaluation frameworks and therefore persist unaddressed over time.

The concept of Feedback-Induced Distribution Shift (FIDS) is introduced to formalize this phenomenon. The three subtypes, Aggregate Masking Effect (AME), Low-Resource Locale Collapse (LRLC), and Aspirational Deferral Lock (ADL), capture distinct but interacting mechanisms through which performance disparities emerge, compound, and remain unresolved in practice. Each is given a minimal formal model with a short proof, and AME and the resource-tier gap are illustrated with public data. Together, they provide a structured lens for understanding how system-level, feedback-driven dynamics can produce uneven outcomes across linguistic populations.

RL-Stable Deployment (RLSD) is proposed as a complementary mitigation framework that operates at inference time to detect and respond to these dynamics. It remains a design, with a conjectured bound. While not a substitute for improved organizational practices or evaluation standards, it offers a practical mechanism for reducing the impact of FIDS in deployed systems.

More broadly, this work highlights a fundamental limitation in how multilingual AI systems are currently evaluated. Metrics that emphasize aggregate performance can obscure meaningful disparities in user experience, particularly in low-resource or underrepresented contexts. As a result, systems may be considered successful despite systematically underperforming for segments of the populations they serve.

Addressing this challenge requires a shift in both measurement and system design. Evaluation frameworks must move beyond aggregate summaries to account for performance across user groups, including worst-case outcomes. Deployment architectures must incorporate mechanisms for detecting and correcting disparities as they emerge. Ultimately, improving multilingual AI quality depends not only on better models, but on better alignment between what systems optimize and what users actually experience.

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Revision note

Version 3 (November 2026) corrects Proposition 2 (clipping to [0,1]), the alert condition in Proposition 1, and the scope of Proposition 3; scopes the no-retraining claim to AME and ADL; adds related work on resource tiers, subgroup audits, and performative prediction; adds an empirical illustration on public data (Section 6); and states the RLSD bound as a conjecture with a toy special case.